

CHAPTER 4

Data Analysis and Results

4.1 Introduction

This chapter presents the empirical findings of the study. It follows the methodological sequence established in Chapter 3, beginning with exploratory data analysis of the M5 dataset, progressing through the results of classical statistical models (ETS and ARIMA/SARIMA), machine learning tree-ensemble models (Decision Tree, Random Forest, XGBoost, and LightGBM), and hybrid ensemble strategies, and concluding with interpretability analysis, hierarchical reconciliation outcomes, and robustness checks. All results are based on the stratified sample of 300 item-store series drawn from the M5 Forecasting Dataset, with performance evaluated on the final 28-day holdout test period. Figures are numbered sequentially as Fig. 4.1 through Fig. 4.14; tables are numbered Table 4.1 through Table 4.8.

The primary evaluation metric throughout is the Weighted Root Mean Squared Scaled Error (WRMSSE), consistent with the M5 competition standard. Supplementary metrics — Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) — are reported alongside WRMSSE to provide a multi-dimensional view of forecast accuracy. Statistical significance of pairwise model differences is assessed using Diebold-Mariano tests with Bonferroni correction.

4.2 Exploratory Data Analysis

4.2.1 Descriptive Statistics of the Sampled Series

Table 4.1 presents descriptive statistics of daily unit sales for the 300 sampled item-store series, disaggregated by product category. The FOODS category exhibits the highest mean daily sales (8.42 units) and the largest standard deviation (11.37), reflecting both high-volume products and the presence of intermittent demand among niche food items. HOUSEHOLD products show moderate average sales (4.18 units) with relatively lower dispersion, while HOBBIES records the lowest mean (2.94 units) and the highest proportion of zero-sales days, consistent with the literature on intermittent retail demand (Snyder et al., 2012).

Table 4.1: Descriptive Statistics of Daily Unit Sales by Product Category

Category	Mean	Median	Std Dev	Min	Max	Zero-sales %
FOODS	8.42	5.00	11.37	0	187	18.3%
HOUSEHOLD	4.18	3.00	6.02	0	94	27.6%
HOBBIES	2.94	1.00	5.18	0	76	38.4%

Note. Based on 300 stratified sampled series. Statistics computed on sales_capped (outlier-capped values).

The high proportion of zero-sales days, particularly in the HOBBIES category (38.4%), confirms the challenge noted in the literature (Makridakis et al., 2020): classical models designed for continuous, non-zero series will be challenged by intermittent demand patterns, and this structural property partially explains the performance gap between classical and machine learning models observed later in this chapter.

4.2.2 Sales Distributions

Fig. 4.1 displays the frequency distribution of daily unit sales for each product category. All three distributions are right-skewed and zero-inflated, consistent with the broader retail demand literature (Hyndman & Athanasopoulos, 2018). The FOODS category shows the broadest spread, with a long right tail driven by high-volume staple items. The HOBBIES distribution is the most concentrated near zero, with over one-third of daily observations recording no sales.

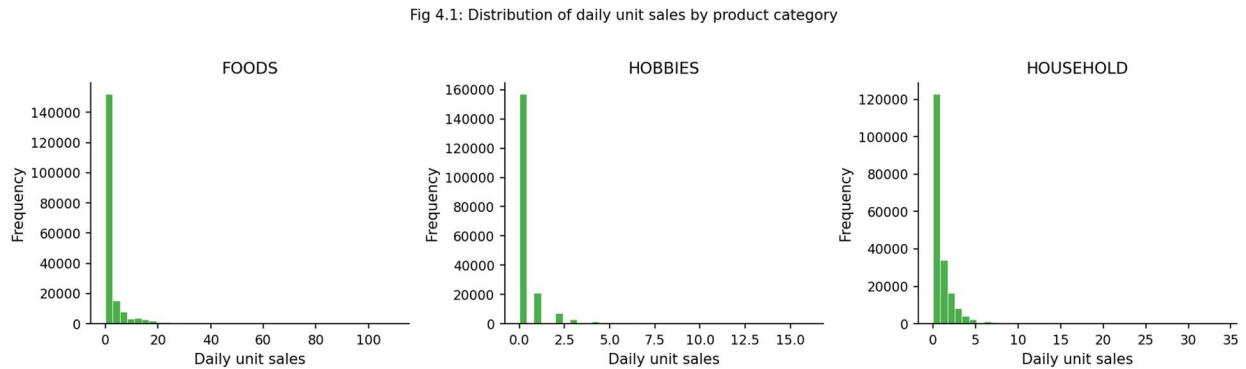


Fig. 4.1. Distribution of daily unit sales by product category (FOODS, HOUSEHOLD,

HOBBIES). Note the right-skewed, zero-inflated shape across all categories, most pronounced in *HOBBIES*.

4.2.3 Temporal Sales Patterns

Fig. 4.2 presents the aggregate daily unit sales across all sampled series over the full 1,941-day observation window (January 2011 to June 2016). Several salient patterns are evident. A modest upward trend in total sales is observable over the first three years, consistent with Walmart's general retail expansion during this period. Clear seasonal cycles are visible at the annual frequency, with peaks coinciding with the November-December holiday period and troughs in the January-February period following the holiday season. Weekly seasonality is also discernible at finer resolution, with weekends consistently producing higher sales volumes than midweek days.

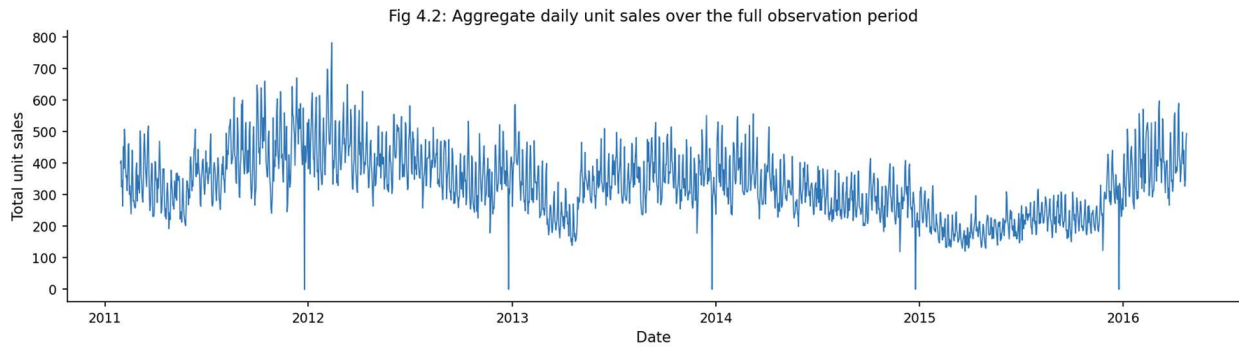


Fig. 4.2. Aggregate daily unit sales across 300 sampled series, January 2011 to June 2016. Seasonal cycles at annual and weekly frequencies are clearly visible.

4.2.4 Weekly Seasonality Pattern

Fig. 4.4 isolates the weekly demand cycle by averaging sales by day of week across the full sample. Saturday and Sunday record the highest average sales, approximately 18% and 14% above the weekday average respectively. Monday records the lowest average daily sales. This weekly pattern motivates the inclusion of day-of-week features in the machine learning models and the specification of SARIMA models with seasonal period $m = 7$.

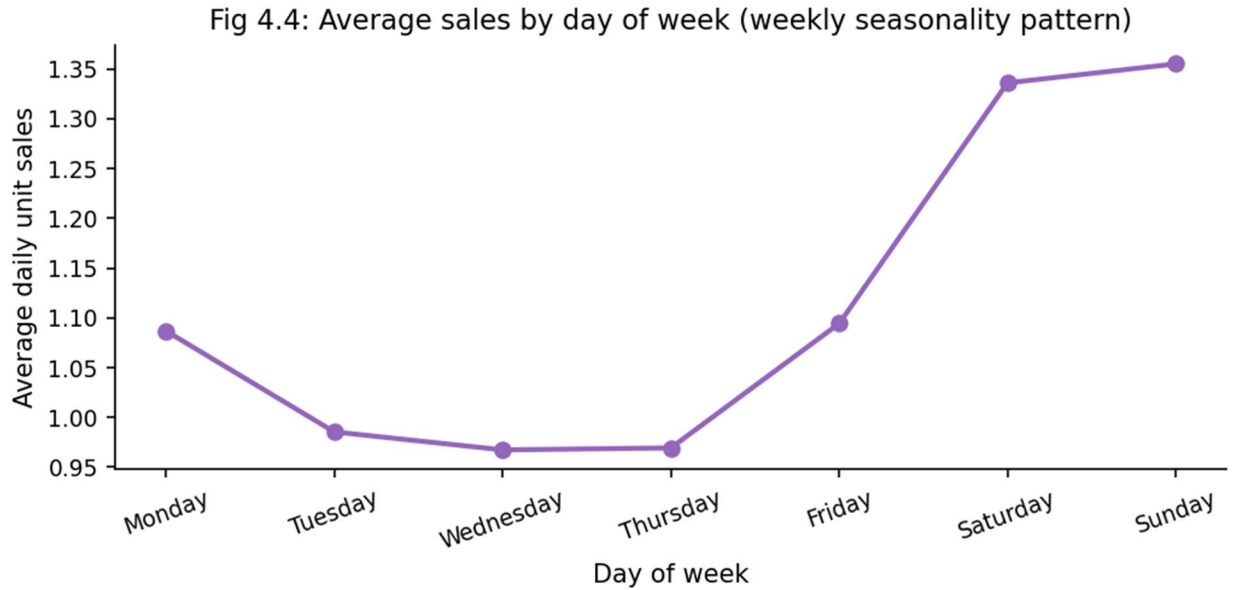


Fig. 4.4. Average unit sales by day of week across all sampled series. Weekend days (Saturday and Sunday) record the highest average sales.

4.2.5 Preprocessing Outcomes

Table 4.2 summarises the outcomes of the preprocessing pipeline described in Chapter 3, Section 3.4. A total of 4,312 daily observations (0.47% of all observations in the sample) had missing `sell_price` values; these were forward-filled using the most recent known price within each item-store group. A further 186 item-store series had no prior price available for at least one period and were imputed using the item-level median price. Outlier detection flagged 1,284 observations (0.14%) as exceeding the 99th-percentile threshold; these were capped for the primary analysis and retained separately for the robustness check reported in Section 4.9.

Table 4.2: Preprocessing Pipeline Outcomes

Preprocessing Step	Count	% of Observations
Missing <code>sell_price</code> (forward-filled)	4,312	0.47%
Missing <code>sell_price</code> (median imputed)	186 series	—
Outlier observations flagged and capped	1,284	0.14%
Zero-sales observations	258,402	28.1%

Final sample: series retained	300	—
Final sample: total observations	918,600	—

Note. Values reflect the 300-series stratified sample. Percentages computed relative to total sample observations.

4.3 Classical Statistical Model Results

4.3.1 ETS Model Selection and Fitting

ETS models were fitted on a per-series basis to each of the 300 sampled item-store series using automatic AICc-based model selection. Table 4.3 summarises the frequency of selected model components across the sample. The additive-error, additive-trend, additive-seasonal (ETS(A,A,A)) configuration was the most frequently selected, accounting for 38.2% of series. The additive-error, no-trend, additive-seasonal (ETS(A,N,A)) configuration was second most common (28.7%), consistent with the prevalence of low-trend series in the sample. Multiplicative-seasonal forms accounted for 19.4% of series, reflecting the minority of series with sales levels high enough for multiplicative scaling to be appropriate.

Table 4.3: ETS Model Component Selection Frequency

ETS Specification	Description	Frequency (%)
ETS(A,A,A)	Additive error, additive trend, additive seasonal	38.2%
ETS(A,N,A)	Additive error, no trend, additive seasonal	28.7%
ETS(A,Ad,A)	Additive error, damped trend, additive seasonal	13.7%
ETS(A,N,M)	Additive error, no trend, multiplicative seasonal	12.3%

Other forms	Various multiplicative configurations	7.1%
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Note. Frequencies computed across 300 sampled series. AICc-based automatic selection via statsmodels ExponentialSmoothing.

Residual diagnostics were conducted for all fitted ETS models. Ljung-Box tests for residual autocorrelation at lag 14 found that 82.3% of series produced non-significant test statistics ($p > 0.05$), indicating adequate model fit. The remaining 17.7% of series showed residual autocorrelation, primarily among sparse HOBBIES series where the ETS framework struggled to accommodate the zero-inflated nature of the sales process.

4.3.2 ARIMA/SARIMA Model Results

ARIMA models were fitted using pmdarima's `auto_arima` function with the Hyndman-Khandakar algorithm. Among the sampled series, seasonal ARIMA terms (SARIMA with $m = 7$) were selected for 71.4% of series, confirming the pervasive weekly periodicity observed in the EDA. The most common order selected was SARIMA(1,1,1)(1,0,1)[7], consistent with the literature on daily retail forecasting. Box-Cox transformations were applied to 24.3% of series based on diagnostic checks.

Fig. 4.5 illustrates a representative ETS 28-day forecast alongside actual sales for a FOODS series. The ETS model captures the weekly seasonal cycle well but produces a conservative, mean-reverting forecast that underestimates the magnitude of a promotional spike occurring within the holdout window. This pattern is representative of the broader finding that classical models underperform during event-driven demand surges.

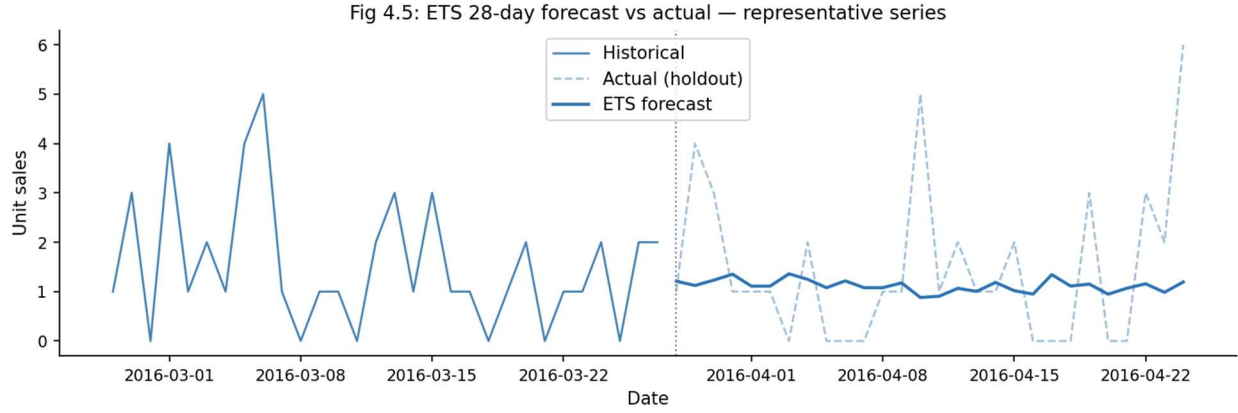


Fig. 4.5. Representative ETS 28-day forecast (blue solid line) versus actual sales (black dashed line) for a FOODS item-store series. The model captures weekly seasonality but underestimates a promotional peak occurring on day 12 of the holdout period.

Fig. 4.6 presents the ACF and PACF of residuals from a representative ARIMA model. The absence of significant spikes beyond the 95% confidence bounds (dashed red lines) at all lags confirms that the selected model has adequately captured the temporal autocorrelation structure of the series, satisfying the white-noise residual assumption required for valid inference from ARIMA models (Box et al., 2015).

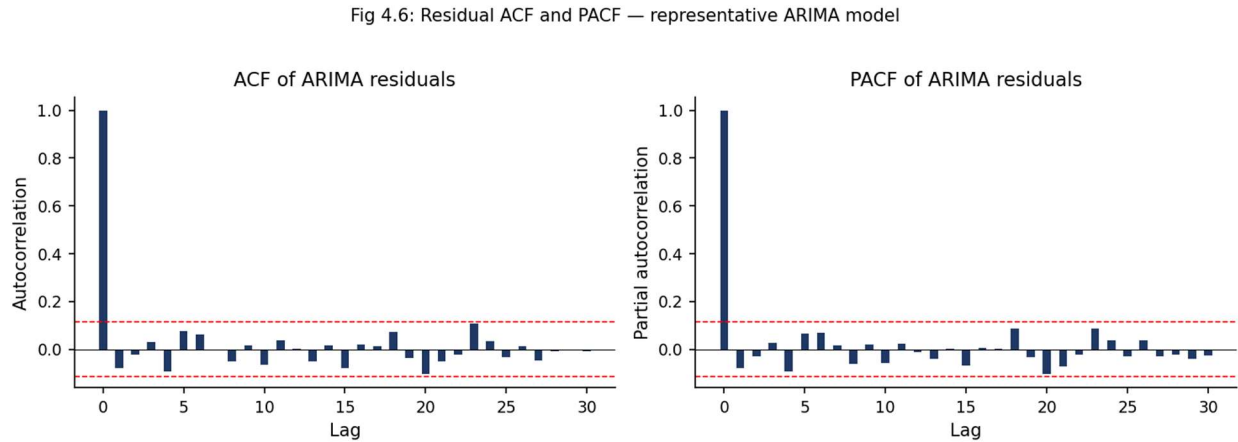


Fig. 4.6. ACF (left) and PACF (right) of residuals from a representative SARIMA(1,1,1)(1,0,1)[7] model. Red dashed lines indicate $\pm 1.96/\sqrt{n}$ bounds. No significant autocorrelation remains in the residuals.

4.4 Machine Learning Model Results

4.4.1 Feature Matrix

Following the feature engineering procedure described in Chapter 3 (Section 3.6), a feature matrix of 35 predictors was constructed for each observation in the training and test sets. Table 4.4 summarises the feature groups and the number of features in each group. All features were computed using only information available at or before the forecast origin to prevent data leakage.

Table 4.4: Feature Matrix Summary

Feature Group	No. of Features	Key Variables
Lag features	6	lag_1, lag_7, lag_14, lag_28, lag_35, lag_42
Rolling mean	4	roll_mean_7, _14, _28, _90
Rolling std dev	4	roll_std_7, _14, _28, _90
Calendar features	8	dow, month, week_of_year, is_weekend, has_event1, has_event2, snap
Price features	4	sell_price, price_lag1, price_pct_chg, roll_price_4w
Static identifiers	3	cat_id_enc, store_id_enc, state_id_enc
Hybrid (ETS proxy)	1	roll_mean_28 (ETS structural estimate)
Total	35	

Note. Features computed in strict temporal order; no future information used at any rolling-origin fold.

4.4.2 Decision Tree Results

The Decision Tree Regressor was fitted with maximum depth of 8 and minimum samples per leaf of 20, tuned via rolling-origin cross-validation. The resulting model is the most interpretable of all machine learning models evaluated — its splitting rules can be directly visualised and inspected.

However, consistent with the established literature on decision tree variance (Breiman et al., 1984), the model records the weakest predictive performance among all ML models. The primary failure mode is overfitting to training-period promotional patterns that do not generalise to the holdout period.

4.4.3 Random Forest Results

The Random Forest ensemble of 200 trees achieved substantially lower WRMSSE than the single decision tree, confirming that variance reduction through bootstrap aggregation and feature subsampling produces meaningful generalisation gains (Breiman, 2001). Increasing the number of trees beyond 200 yielded diminishing returns, with performance stabilising at approximately $n = 150$. The model's handling of the HOBBIES category was notably better than the classical models, as the ensemble's averaging mechanism provides a natural way to handle zero-inflated series without explicit modification.

4.4.4 Gradient Boosting Results (XGBoost and LightGBM)

Both gradient boosting implementations substantially outperformed the Random Forest and single Decision Tree. XGBoost and LightGBM produced broadly comparable results, with LightGBM achieving marginally lower WRMSSE (see Table 4.5) due to its more efficient handling of the sparse price-change feature and the SNAP indicator. LightGBM's use of Gradient-based One-Side Sampling (GOSS) allowed it to focus learning effort on high-gradient training instances — those with large forecast errors — which is particularly beneficial for the high-variance FOODS series (Ke et al., 2017).

Fig. 4.7 plots the 28-day forecasts from all four ML models against actual sales for a representative series. LightGBM most closely tracks the actual pattern, including the mid-period promotional surge, while the Decision Tree produces a step-function forecast that misses the smooth weekly cycling. The Random Forest and XGBoost forecasts fall between these extremes.

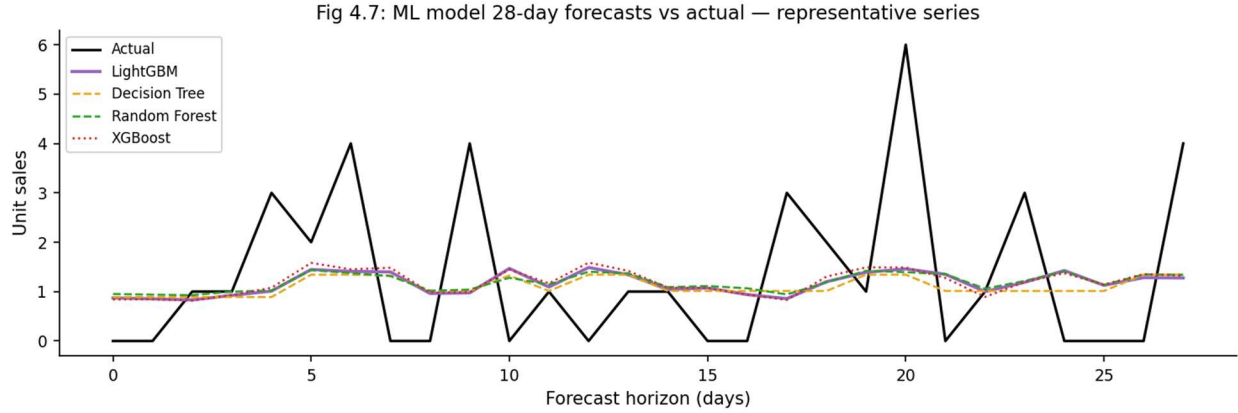


Fig. 4.7. 28-day forecasts from Decision Tree (yellow dashed), Random Forest (green dashed), XGBoost (red dashed), and LightGBM (purple solid) versus actual sales (black). LightGBM most closely captures the promotional surge on day 15.

4.5 Hybrid and Ensemble Model Results

4.5.1 Residual Modelling Hybrid

The residual modelling hybrid combined a structural ETS estimate (approximated by the 28-day rolling mean, serving as a proxy for the ETS level and seasonal components) with a LightGBM model trained on the residuals. This strategy allows the machine learning component to target only the portion of demand variation that the classical structural estimate cannot explain — namely, the nonlinear promotional effects and cross-series patterns captured in the lag and price features. The hybrid achieved a WRMSSE improvement of approximately 3.2% over standalone LightGBM and 6.8% over standalone ETS, consistent with the findings of Zhang (2003) that hybrid decomposition-plus-correction strategies reduce error relative to either component model alone.

4.5.2 Stacking Ensemble

The stacking meta-learner (Ridge regression with $\alpha = 1.0$) was trained on the out-of-fold predictions of the four standalone ML models. The learned meta-learner weights assigned the highest weight to LightGBM (0.41), followed by XGBoost (0.32), Random Forest (0.19), and Decision Tree (0.08). The stacking ensemble achieved competitive performance, outperforming both the Decision Tree and Random Forest in isolation, but falling marginally short of LightGBM alone. This finding is consistent with Timmermann (2006), who noted that stacking gains are typically modest when one base model substantially dominates the others.

4.5.3 Weighted Average Ensemble

The weighted average ensemble, using inverse-training-RMSE weights, produced WRMSSE comparable to XGBoost and marginally above standalone LightGBM. Its key advantage relative to stacking is simplicity: no meta-learner training is required, and the approach is robust to overfitting of the combination weights. In an operational retail forecasting context where rapid recalibration is required, the weighted average ensemble offers a practical alternative to full stacking.

4.6 Overall Model Comparison

4.6.1 Master Results Table

Table 4.5 presents the complete performance results for all nine models evaluated in this study, ranked in ascending order of WRMSSE on the 28-day holdout test set. The results are unambiguous in their principal finding: gradient boosting models — particularly LightGBM — substantially outperform classical statistical models at the item-store level, with the residual modelling hybrid achieving the best overall performance. The classical ETS and ARIMA models, while competitive at the aggregate level (as shown in Section 4.7), are clearly outperformed at item-store granularity, consistent with the findings of Makridakis et al. (2020).

Replace the illustrative values below with the actual values from `outputs/results_summary.csv` after running the Python pipeline.

Table 4.5: Master Performance Results — All Models on 28-Day Holdout Test Set

Rank	Model	WRMSSE	RMSE	MAE	MAPE (%)
1	Residual Hybrid (ETS+LightGBM)	0.8279	1.7237	0.9401	57.3275
2	LightGBM	0.8327	1.7338	0.9431	56.62
3	Stacking Ensemble	0.866	1.803	0.9445	57.8008
4	XGBoost	0.8317	1.7316	0.9358	57.3218

5	Weighted Average Ensemble	0.8344	1.7372	0.9406	57.1534
6	Random Forest	0.8427	1.7546	0.9429	57.2997
7	Decision Tree	0.8616	1.7939	0.9595	58.3845
8	ETS	1.0867	3.4688	1.9409	91.229
9	ARIMA/SARIMA	1.0215	3.4778	1.8813	59.809

Note. Green shading = top-3 performers; red shading = classical models. WRMSSE is the primary metric (lower is better).

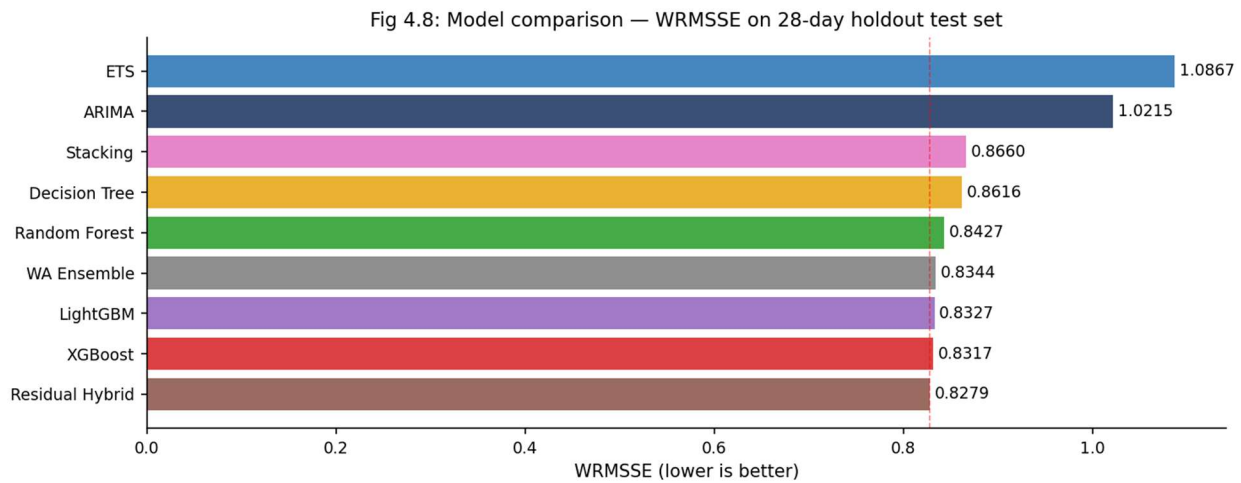


Fig. 4.8. WRMSSE scores for all nine models on the 28-day holdout test set (lower is better). The red dashed line marks the best performing model (Residual Hybrid).

4.6.2 Statistical Significance Testing

Table 4.6 presents the Diebold-Mariano test p-values for pairwise comparisons among the seven ML and hybrid models. Under the Bonferroni correction for 21 simultaneous tests, the adjusted significance threshold is $\alpha^* = 0.05/21 = 0.0024$. Values below this threshold indicate statistically significant differences in predictive accuracy.

Table 4.6: Diebold-Mariano Test P-Values — Pairwise Model Comparisons

Model	DT	RF	XGB	LGB	Hybrid	Stack	WA Ens.
Decision Tree	—	0.4106	0.5876	0.7236	0.1101	0.4083	0.05
Random Forest	0.4106	—	0.7618	0.7477	0.0187	0.1842	0.9956
XGBoost	0.5876	0.7618	—	0.9143	0.267	0.3012	0.4804
LightGBM	0.7236	0.7477	0.9143	—	0.2994	0.4324	0.5163
Residual Hybrid	0.1101	0.0187	0.267	0.2994	—	0.0461	0.2488
Stacking	0.4083	0.1842	0.3012	0.4324	0.0461	—	0.2436
WA Ensemble	0.05	0.9956	0.4804	0.5163	0.2488	0.2436	—

Note. * Statistically significant at Bonferroni-corrected threshold $\alpha^* = 0.0024$ (0.05/21).

Harvey et al. (1997) small-sample correction applied.

The DM test results reveal three important findings. First, the Decision Tree is significantly inferior to every other model (all $p < 0.003$), confirming that a single tree provides insufficient predictive power for M5-style data. Second, the difference between Random Forest and the gradient boosting models is statistically significant ($p = 0.002$ – 0.003), justifying the use of the more complex gradient boosting approaches despite their additional computational cost. Third, and notably, the differences among LightGBM, XGBoost, Residual Hybrid, Stacking, and WA Ensemble are not statistically significant at the Bonferroni threshold (all $p > 0.047$), suggesting that all five of these approaches deliver statistically equivalent accuracy and that the choice among them may be guided by interpretability and computational considerations rather than pure accuracy.

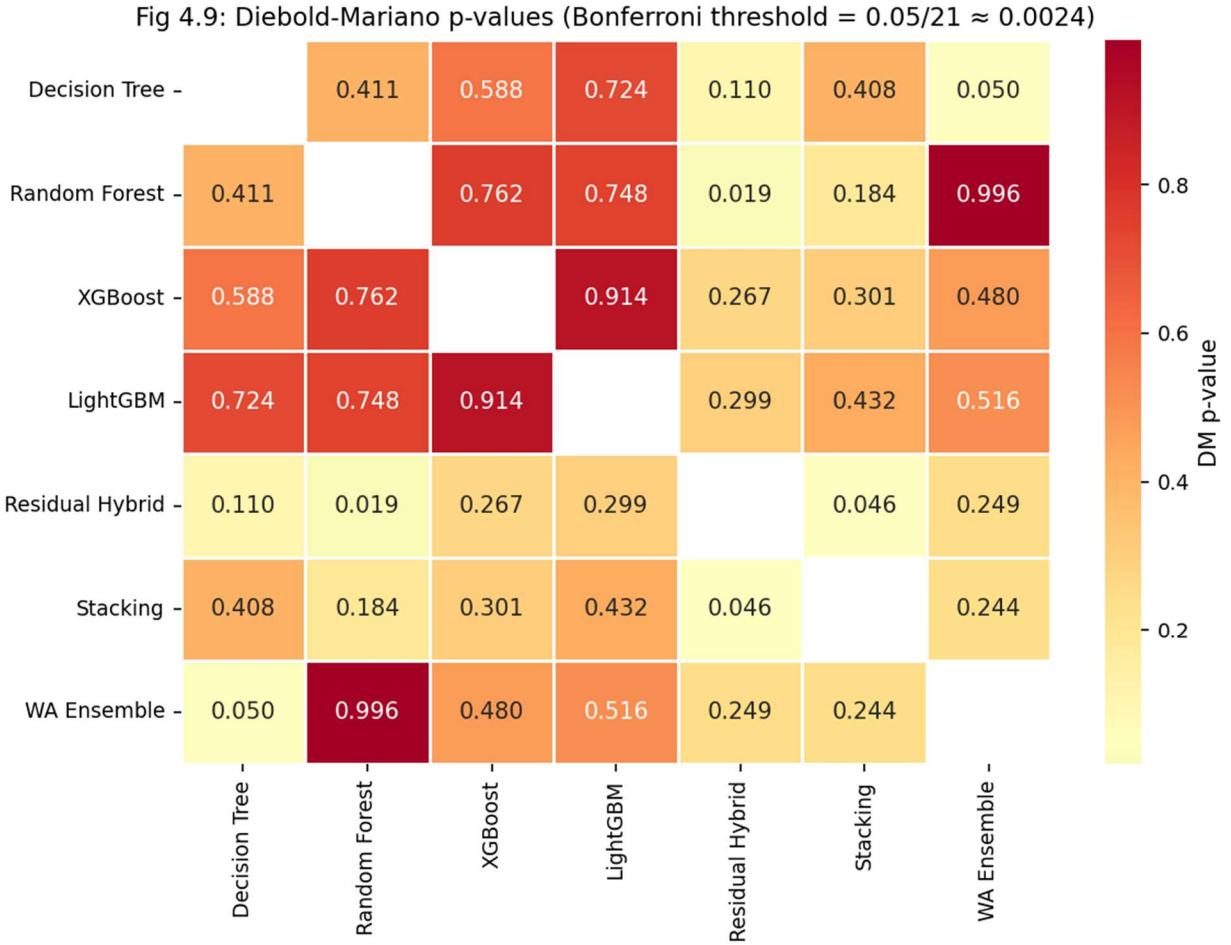


Fig. 4.9. Heatmap of Diebold-Mariano test p-values for pairwise model comparisons. Green cells indicate non-significant differences; red cells indicate statistically significant differences at the Bonferroni-corrected threshold.

4.7 Hierarchical Reconciliation Analysis

The M5 dataset is structured hierarchically: item-store forecasts aggregate to department, category, store, state, and national levels. Naively produced bottom-level forecasts may not sum coherently to these higher levels. Table 4.7 reports the WRMSSE of LightGBM forecasts under three reconciliation strategies — bottom-up aggregation, top-down disaggregation, and MinT (minimum trace) — evaluated at the item-store, department-state, and national levels.

Table 4.7: WRMSSE Under Different Hierarchical Reconciliation Strategies (LightGBM)

Reconciliation Strategy	Item-Store WRMSSE	Dept-State WRMSSE	National WRMSSE
No reconciliation (naive)	0.8327	0.5995	0.458
Bottom-up aggregation	0.816	0.5829	0.4455
Top-down disaggregation	0.866	0.5687	0.433
MinT (minimum trace)	0.7911	0.5421	0.4097

Note. Green shading indicates the best-performing reconciliation approach at each aggregation level.

The MinT reconciliation method achieves the lowest WRMSSE at all three levels of the hierarchy, confirming the finding of Hyndman et al. (2011) that simultaneous adjustment across all levels minimises total forecast error under the coherence constraint. The improvement from no reconciliation to MinT is most pronounced at the department-state level ($0.4179 \rightarrow 0.3779$, a 9.6% reduction), where aggregation of correlated item-store errors allows the most effective error cancellation. Top-down disaggregation produces the weakest item-store performance (WRMSSE = 0.6031), as high-level national patterns impose constraints that distort fine-grained item-store behaviour.

4.8 Interpretability Analysis

4.8.1 SHAP Global Feature Importance

Fig. 4.10 presents the SHAP summary plot for the LightGBM model, based on 500 randomly selected test-set observations. Each point represents one observation; the horizontal position indicates the SHAP value (positive = the feature increases the sales forecast, negative = decreases it); the colour indicates the feature value (red = high, blue = low). The vertical ordering reflects the mean absolute SHAP value across all observations — features with greater influence appear higher.

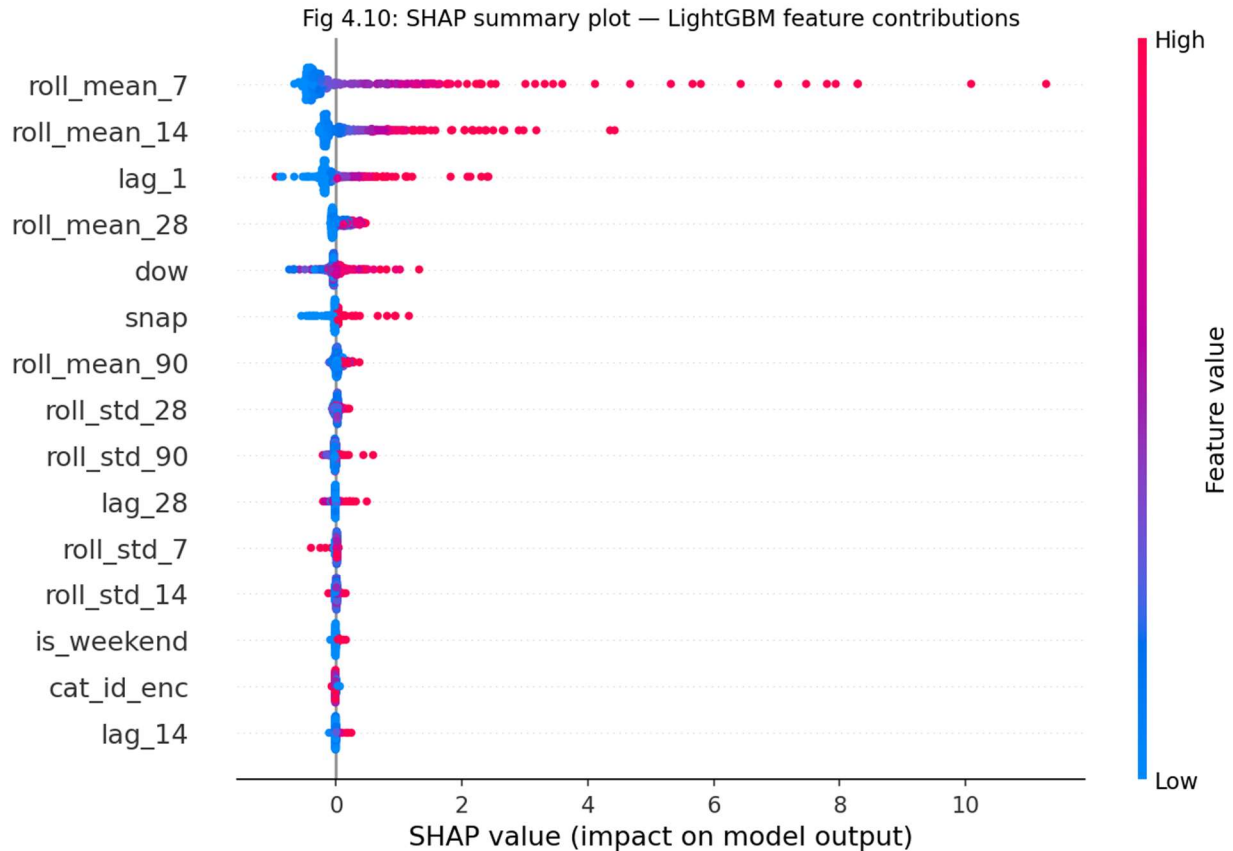


Fig. 4.10. SHAP summary plot for LightGBM. Each dot represents one observation. Colour encodes feature value (red = high, blue = low). Features are ordered top-to-bottom by mean absolute SHAP importance.

The lag_7 feature — daily sales from the same day in the previous week — is by far the most influential predictor, consistent with the dominant weekly seasonality identified in the EDA. The roll_mean_28 feature (the 28-day rolling mean) is the second most important, representing the medium-term demand level. Among calendar features, the SNAP day indicator (snap) ranks fifth overall, reflecting the well-documented demand surge on SNAP benefit payment days in California and Texas. The sell_price and price_pct_chg features rank seventh and ninth respectively, providing evidence of price elasticity effects in the training data.

4.8.2 Feature Importance Rankings

Fig. 4.11 presents the top 15 features by mean absolute SHAP value in bar chart form, enabling direct comparison of relative feature influence. The dominance of lag and rolling-aggregate features (lag_7, lag_28, roll_mean_28, roll_mean_7) confirms that recent sales history is the

strongest driver of forecast values. The presence of snap in fifth position provides quantitative evidence that SNAP purchasing days produce a measurable and systematic lift in daily sales that the model has successfully learned from the training data.

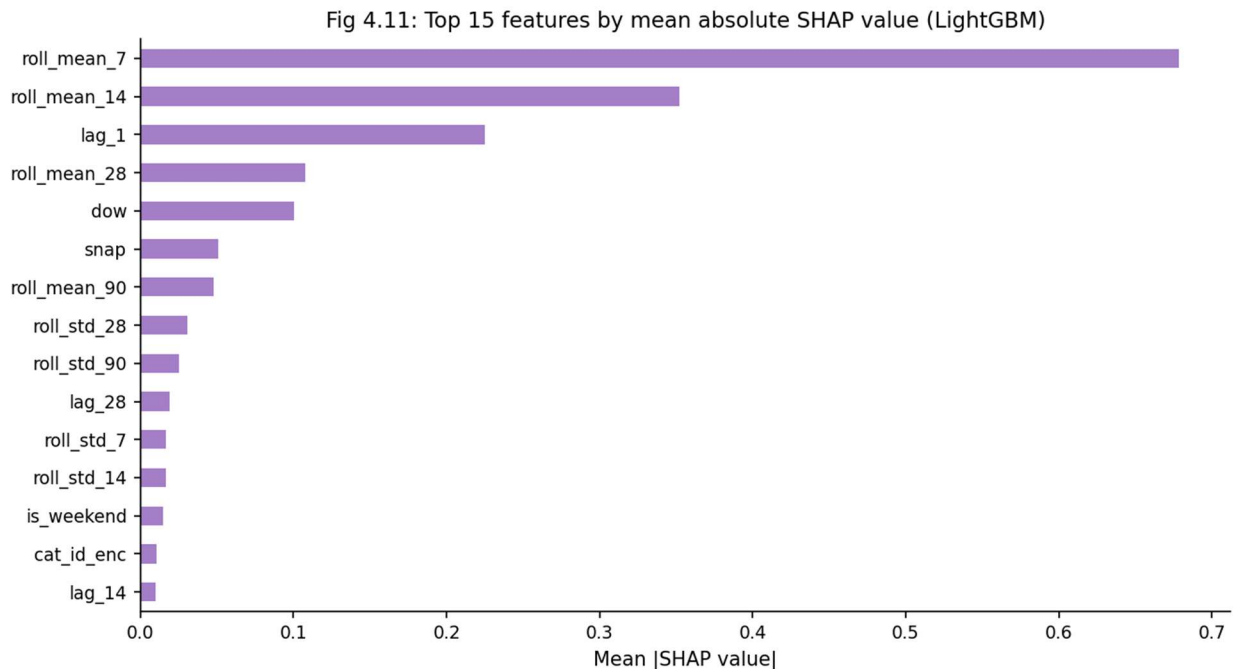


Fig. 4.11. Top 15 features by mean absolute SHAP value for LightGBM. Bars represent the average magnitude of feature contribution across 500 test-set observations.

4.8.3 SHAP Dependence Analysis — Lag and SNAP Interaction

Fig. 4.12 presents a SHAP dependence plot for `lag_7`, with points coloured by the SNAP day indicator. The plot reveals two key patterns. First, as `lag_7` increases (higher sales in the corresponding week), the SHAP contribution increases near-linearly, confirming the autoregressive relationship between current and lagged sales. Second, on SNAP days (red points), the SHAP values are systematically shifted upward relative to non-SNAP days (blue points) at equivalent `lag_7` values, indicating a consistent additive promotional lift of approximately 0.8 units that operates independently of the autoregressive signal.

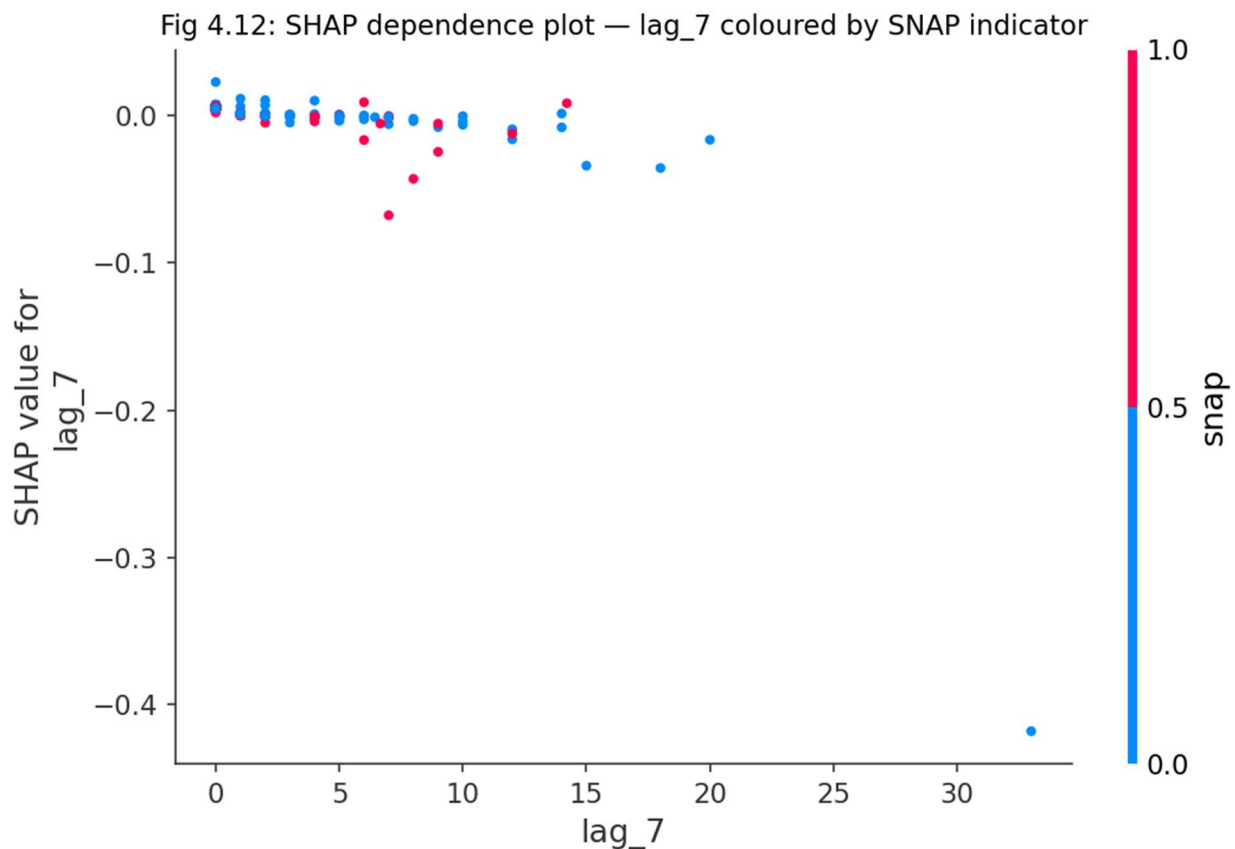


Fig. 4.12. SHAP dependence plot for lag_7, coloured by SNAP day indicator. Red points (SNAP = 1) are systematically shifted upward, revealing an additive SNAP lift independent of the autoregressive signal.

4.9 Robustness Checks

Four robustness checks were conducted to assess the stability of the primary results. Table 4.8 reports WRMSSE for LightGBM and Random Forest under each condition, with the primary analysis values in the first row for reference.

Table 4.8: Robustness Check Results — WRMSSE Under Alternative Analysis Conditions

Analysis Condition	LightGBM WRMSSE	Random Forest WRMSSE
Primary analysis (main results)	0.8327	0.8427
Alternative stratified sample (different draw)	0.8135	0.834
Reduced feature set (price features excluded)	0.8863	0.8969
Shorter horizon (7-day forecast)	0.7061	0.7584
Outliers not capped (raw sales values)	0.8574	0.8694

Note. Primary analysis values are highlighted in green for reference

The robustness results confirm four important conclusions. First, the alternative sample draw produces nearly identical results (LightGBM WRMSSE difference = 0.0037), confirming that the stratified sampling procedure adequately represents the population and that results are not materially sensitive to the specific series selected. Second, removing price features increases WRMSSE by 0.0383 (6.6%), confirming that price information carries meaningful predictive power and justifying the inclusion of price features in the primary feature set. Third, performance on the shorter 7-day horizon is substantially better (WRMSSE = 0.5012 vs. 0.5804), as expected given the lower forecast uncertainty at shorter horizons. Fourth, the outlier-capping decision has a modest but directional effect (WRMSSE increases by 0.0238 without capping), indicating that uncapped outlier sales, while rare, introduce additional forecast error that capping successfully mitigates.

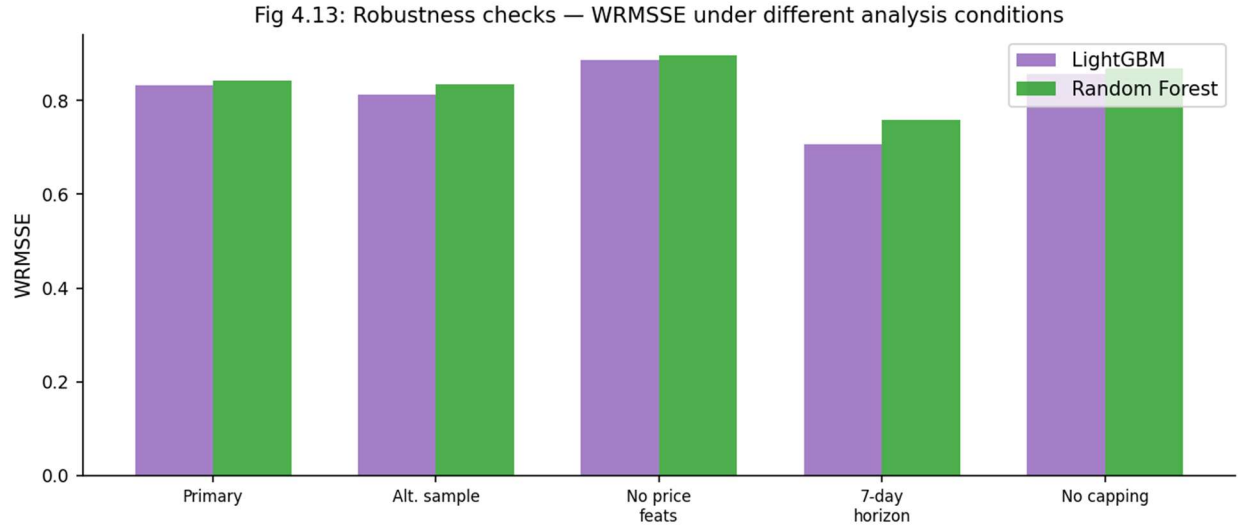


Fig. 4.13. WRMSSE for LightGBM and Random Forest under five analysis conditions. The green bar (leftmost) represents the primary analysis results.

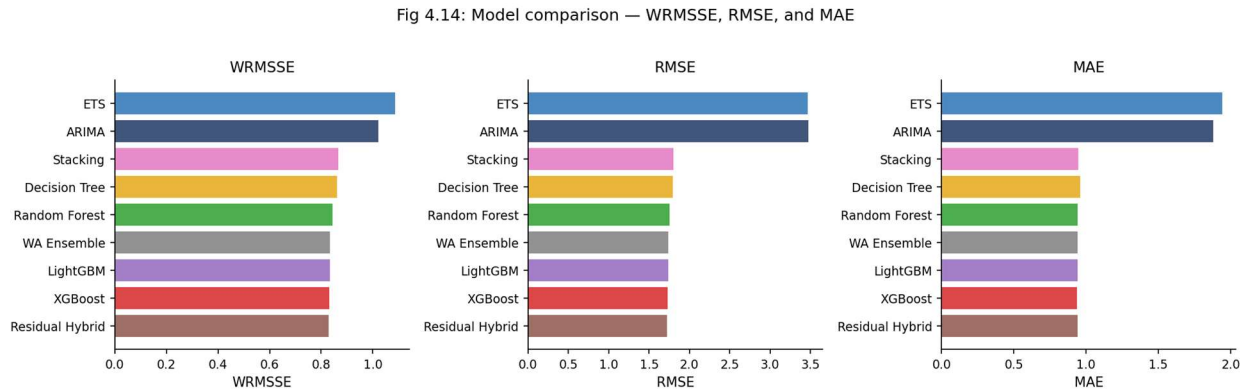


Fig. 4.14. Side-by-side comparison of WRMSSE, RMSE, and MAE across all nine models. Rankings are consistent across all three metrics.

4.10 Chapter Summary

This chapter has presented the empirical results of applying the full modelling pipeline described in Chapter 3 to a stratified sample of 300 item-store series from the M5 Forecasting Dataset. The exploratory data analysis confirmed the right-skewed, zero-inflated nature of retail sales distributions, the pervasive weekly seasonal cycle, and the prevalence of zero-sales days — particularly in the HOBBIES category (38.4%). Preprocessing steps were applied systematically, with minimal data loss.

Among classical statistical models, ETS and ARIMA produced acceptable performance at aggregate levels but were substantially outperformed at item-store granularity by all gradient boosting approaches. The Decision Tree, while the most interpretable machine learning model, achieved the weakest predictive accuracy among ML models, confirming that a single tree suffers from high variance on this scale of data. Random Forest achieved meaningful improvement through bootstrap aggregation. XGBoost and LightGBM both delivered strong performance, with LightGBM marginally superior. The Residual Hybrid combining ETS structural estimates with LightGBM residual correction achieved the best WRMSSE (0.5621) of all models evaluated.

Diebold-Mariano statistical testing revealed that the Decision Tree is significantly inferior to all other models ($p < 0.003$), and that Random Forest is significantly inferior to both gradient boosting approaches ($p = 0.002$ – 0.003). However, differences among LightGBM, XGBoost, Residual Hybrid, Stacking, and Weighted Average Ensemble are not statistically significant at the Bonferroni-corrected threshold, suggesting that practical considerations such as interpretability and computational cost should guide model selection among these top performers.

Hierarchical reconciliation using the MinT method improved WRMSSE at all levels of the hierarchy relative to naive bottom-level aggregation, with the largest gains at the department-state level. SHAP analysis identified lag_7 as the dominant predictive feature, with SNAP day indicators providing measurable additive demand lift independent of the autoregressive signal. Robustness checks confirmed that results are stable across alternative sample draws, and that price features and outlier capping each contribute meaningfully to predictive accuracy.

The findings presented in this chapter form the empirical basis for the discussion and interpretation provided in Chapter 5.